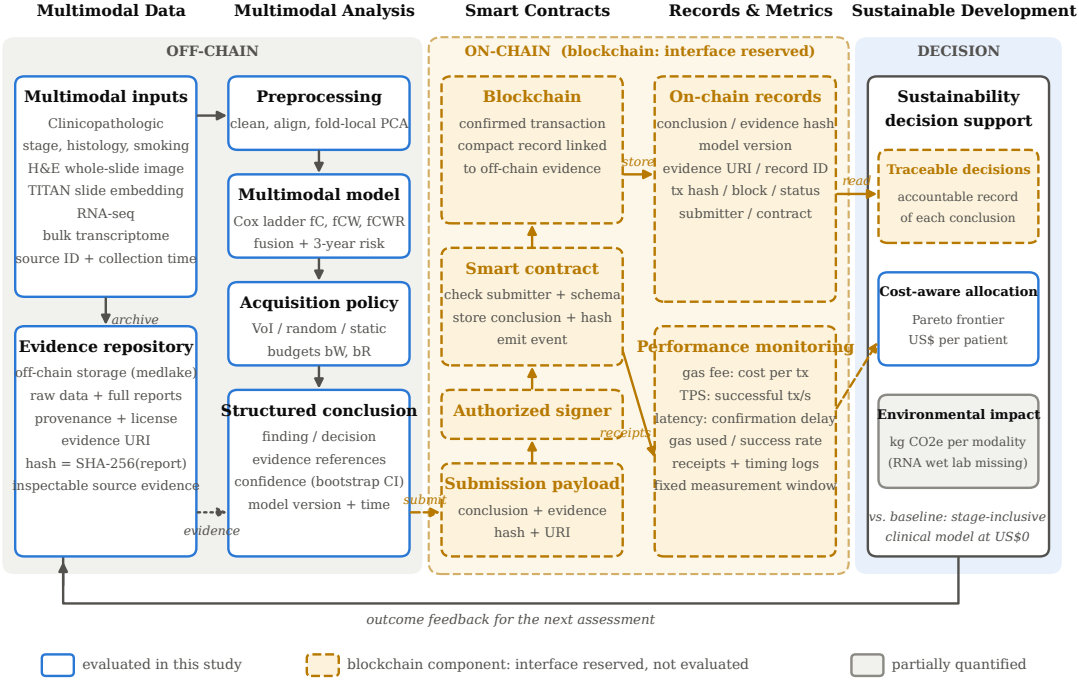
Multimodal Data
Multimodal Analysis
Smart Contracts
Records & Metrics
Sustainable Development
OFF-CHAIN
Multimodal inputs
Clinicopathologic
stage, histology, smoking
H&E whole-slide image
TITAN slide embedding
RNA-seq
bulk transcriptome
source ID + collection time
archive
Evidence repository
off-chain storage (medlake)
raw data + full reports
provenance + license
evidence URI
hash = SHA-256(report)
inspectable source evidence
evidence
Preprocessing
clean, align, fold-local PCA
Multimodal model
Cox ladder fC, fCW, fCWR
fusion + 3-year risk
Acquisition policy
VoI / random / static
budgets bW, bR
Structured conclusion
finding / decision
evidence references
confidence (bootstrap CI)
model version + time
submit
ON-CHAIN (blockchain: interface reserved)
Blockchain
confirmed transaction
compact record linked
to off-chain evidence
store
Smart contract
check submitter + schema
store conclusion + hash
emit event
Authorized signer
Submission payload
conclusion + evidence
hash + URI
receipts
On-chain records
conclusion / evidence hash
model version
evidence URI / record ID
tx hash / block / status
submitter / contract
Performance monitoring
gas fee: cost per tx
TPS: successful tx/s
latency: confirmation delay
gas used / success rate
receipts + timing logs
fixed measurement window
read
DECISION
Sustainability
decision support
Traceable decisions
accountable record
of each conclusion
Cost-aware allocation
Pareto frontier
US$ per patient
Environmental impact
kg CO2e per modality
(RNA wet lab missing)
vs. baseline: stage-inclusive
clinical model at US$0
outcome feedback for the next assessment
evaluated in this study
blockchain component: interface reserved, not evaluated
partially quantified